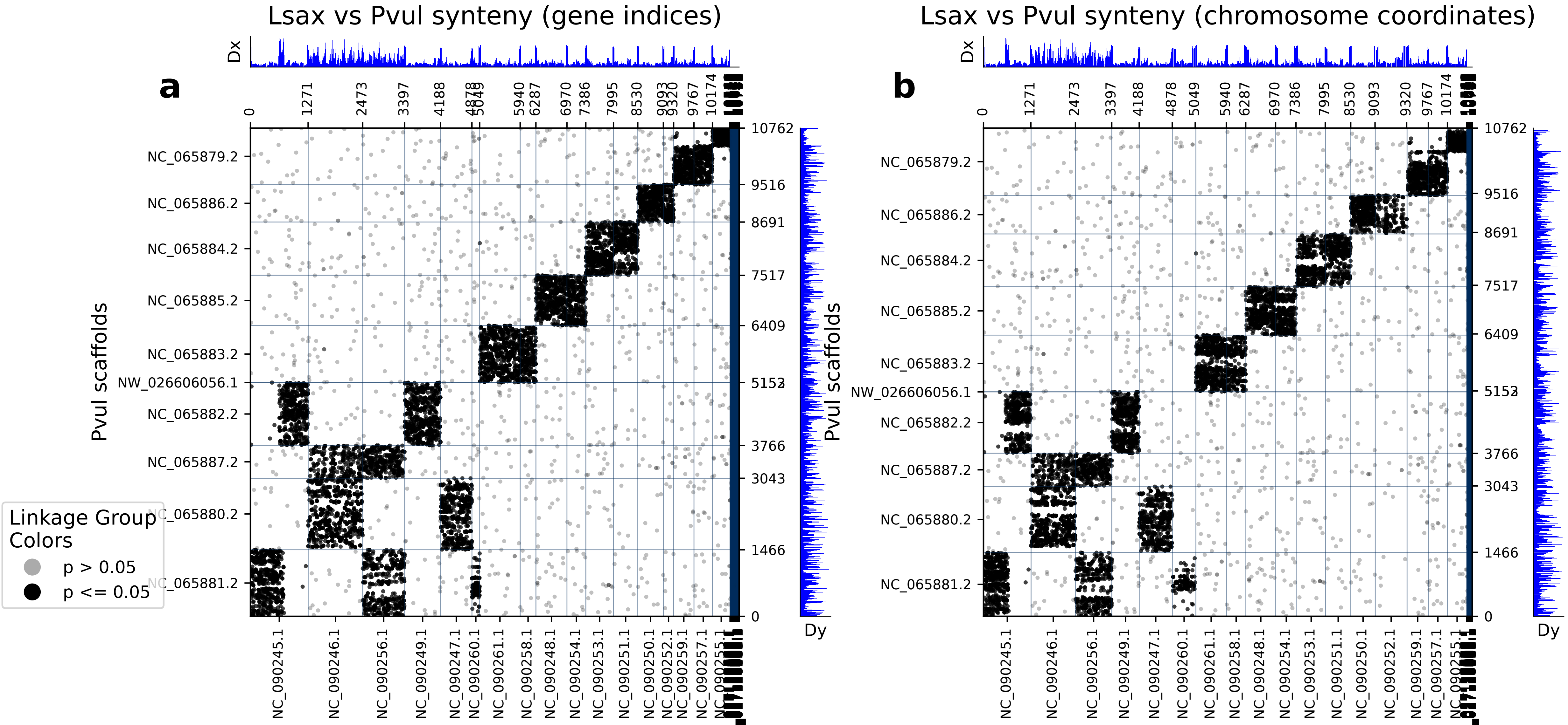
diamond: 10763 orthologs on 192 Lsax scaffolds and 10 Pvul scaffolds.



c Lsax vs Pvul groups

Lsax scaffolds

d Lsax vs Pvul FET, whole chroms.

	Lsax scaf	Pvul scaf	p (FET)	Count
0	NC_090246.1	NC_065880.2	0.00e+00	841
1	NC_090261.1	NC_065883.2	0.00e+00	798
2	NC_090249.1	NC_065882.2	0.00e+00	707
3	NC_090245.1	NC_065881.2	8.33e-258	640
4	NC_090247.1	NC_065880.2	0.00e+00	613
5	NC_090248.1	NC_065885.2	0.00e+00	608
6	NC_090253.1	NC_065884.2	0.00e+00	539
7	NC_090245.1	NC_065882.2	3.10e-175	536
8	NC_090256.1	NC_065881.2	1.35e-213	503
9	NC_090250.1	NC_065886.2	0.00e+00	503
10	NC_090251.1	NC_065884.2	0.00e+00	481
11	NC_090259.1	NC_065879.2	0.00e+00	393
12	NC_090254.1	NC_065885.2	0.00e+00	366
13	NC_090256.1	NC_065887.2	3.08e-209	362
14	NC_090257.1	NC_065879.2	5.76e-293	358
15	NC_090255.1	NC_065879.2	2.28e-250	325
16	NC_090258.1	NC_065883.2	3.85e-262	314
17	NC_090246.1	NC_065887.2	1.22e-77	267
18	NC_090252.1	NC_065886.2	4.64e-181	192
19	NC_090260.1	NC_065881.2	4.76e-95	144

e Lsax vs Pvul FET, chrom. pieces

Lsax scaffolds

	Lsax scaf	Pvul scaf	p (FET)	Count
0	NC_090246.1 : 0-end	NC_065880.2 : 0-end	0.00e+00	841
1	NC_090261.1 : 0-end	NC_065883.2 : 0-end	0.00e+00	798
2	NC_090249.1 : 0-end	NC_065882.2 : 0-end	0.00e+00	707
3	NC_090245.1 : 0-end	NC_065881.2 : 0-end	8.33e-258	640
4	NC_090247.1 : 0-end	NC_065880.2 : 0-end	0.00e+00	613
5	NC_090248.1 : 0-end	NC_065885.2 : 0-end	0.00e+00	608
6	NC_090253.1 : 0-end	NC_065884.2 : 0-end	0.00e+00	539
7	NC_090245.1 : 0-end	NC_065882.2 : 0-end	3.10e-175	536
8	NC_090256.1 : 0-end	NC_065881.2 : 0-end	1.35e-213	503
9	NC_090250.1 : 0-end	NC_065886.2 : 0-end	0.00e+00	503
10	NC_090251.1 : 0-end	NC_065884.2 : 0-end	0.00e+00	481
11	NC_090259.1 : 0-end	NC_065879.2 : 0-end	0.00e+00	393
12	NC_090254.1 : 0-end	NC_065885.2 : 0-end	0.00e+00	366
13	NC_090256.1 : 0-end	NC_065887.2 : 0-end	3.08e-209	362
14	NC_090257.1 : 0-end	NC_065879.2 : 0-end	5.76e-293	358
15	NC_090255.1 : 0-end	NC_065879.2 : 0-end	2.28e-250	325
16	NC_090258.1 : 0-end	NC_065883.2 : 0-end	3.85e-262	314
17	NC_090246.1 : 0-end	NC_065887.2 : 0-end	1.22e-77	267
18	NC_090252.1 : 0-end	NC_065886.2 : 0-end	4.64e-181	192
19	NC_090260.1 : 0-end	NC_065881.2 : 0-end	4.76e-95	144

(a) depicts the chromosomes/scaffolds of Lsax (x-axis) plotted against the chromosomes/scaffolds of Pvul (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10763 orthologs found between 192 Lsax scaffolds and 10 Pvul scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.